

Short Report: AO Unit Performance Analysis

Experiment Setup:

- CMOS sensor: 62MP with 9576 x 6388 pixels [*from datasheet of ZWO ASI 6200MM-PRO*]
- Single pixel size: 3.76 µm
- AO unit: Starlight Xpress Active Optics Guiding System without OAG
- Maximum steps: 45 steps (Old AO) and 1000 steps (New AO)

Methodology:

The pixel position of a reference point on the CMOS sensor was measured after moving the AO unit in different directions (West, North, South, East) and returning to the center position each time. The experiments were conducted with both the old and new AO units.

Results and Calculations:

For each experiment, calculated the total displacement using $\text{Displacement} = \sqrt{[(X_{\text{final}} - X_{\text{initial}})^2 + (Y_{\text{final}} - Y_{\text{initial}})^2]}$

Then, converted pixel the displacements to micrometers using the given pixel size of 3.76 µm.

[Please visit, [Appendix A](#) or [Appendix B](#) for details calculation]

Key Findings:

Experiment1		Experiment 2		Experiment 3		Experiment 4	
Direction	Old AO 1 step(µm)	Old AO 10 steps (µm)	Old AO 1 Step (µm/step)	New AO 20 Steps (µm)	New AO 1 Step (µm/step)	New AO 100 Steps (µm)	New AO 1 Step (µm/step)
West	18.00	7.08	0.708	10.11	0.506	32.80	0.328
North	35.24	6.85	0.685	11.58	0.579	38.27	0.383
South	44.22	6.77	0.677	9.58	0.479	33.58	0.336
East	4.64	6.82	0.682	9.89	0.495	35.50	0.355

Experiment 5(Repeated exp4)

Experiment 6

Direction	New AO 100 steps(μm)	New AO 1 step ($\mu\text{m}/\text{step}$)	New AO 1000 Steps (μm)	New AO 1 Step ($\mu\text{m}/\text{step}$)
West	31.01	0.310	35.81	0.0358
North	38.97	0.390	39.78	0.0398
South	34.76	0.348	34.12	0.0341
East	36.41	0.341	36.60	0.0366

Conclusions:

1. The old AO unit shows significant variability in step size, with larger movements in the North-South direction compared to the East-West direction.
2. The new AO unit demonstrates more consistent step sizes across different directions, with smaller movements per step compared to the old AO unit.
3. The variability in return-to-center movements suggests the presence of hysteresis in both AO units. Hysteresis can be minimized by:
 - Implementing a calibration routine to account for and correct the hysteresis
 - Using smaller step sizes and iterating as needed, rather than relying on large single-step movements.
 - Ensuring mechanical stability and minimizing external disturbances such as vibrations and thermal fluctuations.

Appendix A

Below, are the tables of data collected(using Maximdl) during the experiments -

[Old AO Unit]

Experiment 1: 1 Step

Position	X	Y
Centre	4169.111	2641.752
Centre	4168.854	2644.634
West	4168.854	2646.532
Centre	4170.048	2647.622
North	4169.844	2638.253
Centre	4169.567	2626.795
South	4166.804	2615.408
Centre	4166.017	2619.758
East	4166.544	2618.544
Centre	4166.321	2618.926

Experiment 2: 10 Steps

Position	X	Y
Centre	4664.791	3200.447
West	4666.674	3200.521
Centre	4664.968	3200.926
North	4664.696	3199.136
Centre	4664.935	3200.614
South	4664.742	3202.414
Centre	4664.924	3200.193
East	4663.116	3200.045
Centre	4664.959	3199.849
Return	4664.672	3200.487

[New AO Unit]

Experiment 3: 20 Steps

Position	X	Y
Centre	4243.739	3128.057
West	4246.305	3127.305
Centre	4243.68	3127.622
North	4243.253	3124.584
Centre	4243.676	3126.815
South	4244.157	3129.281
Centre	4243.412	3127.37
East	4240.781	3127.318
Centre	4243.053	3126.241
Return	4243.883	3126.767

Experiment 4: 100 Steps

Position	X	Y
Centre	4243.32	3126.364
West	4251.762	3124.839
Centre	4243.51	3126.252
North	4242.112	3116.439
Centre	4242.938	3126.23
South	4244.229	3134.987
Centre	4243.212	3125.807
East	4233.869	3126.851
Centre	4243.736	3125.855
Return	4243.515	3126.234

Appendix B

Calculations

Old AO Unit:

Experiment 1: 1 Step

1. Centre to West:

- $\Delta X = 4168.854 - 4169.111 = -0.257$ pixels
- $\Delta Y = 2646.532 - 2641.752 = 4.78$ pixels
- Movement = $\sqrt{((-0.257 * 3.76)^2 + (4.78 * 3.76)^2)} = 18.00 \mu\text{m}$

2. Centre to North:

- $\Delta X = 4169.844 - 4170.048 = -0.204$ pixels
- $\Delta Y = 2638.253 - 2647.622 = -9.369$ pixels
- Movement = $\sqrt{((-0.204 * 3.76)^2 + (-9.369 * 3.76)^2)} = 35.24 \mu\text{m}$

3. Centre to South:

- $\Delta X = 4166.804 - 4169.567 = -2.763$ pixels
- $\Delta Y = 2615.408 - 2626.795 = -11.387$ pixels
- Movement = $\sqrt{((-2.763 * 3.76)^2 + (-11.387 * 3.76)^2)} = 44.22 \mu\text{m}$

4. Centre to East:

- $\Delta X = 4166.544 - 4166.017 = 0.527$ pixels
- $\Delta Y = 2618.544 - 2619.758 = -1.214$ pixels
- Movement = $\sqrt{((0.527 * 3.76)^2 + (-1.214 * 3.76)^2)} = 4.64 \mu\text{m}$

Experiment 2: 10 Steps

1. Centre to West:

- $\Delta X = 4666.674 - 4664.791 = 1.883$ pixels
- $\Delta Y = 3200.521 - 3200.447 = 0.074$ pixels
- Movement = $\sqrt{((1.883 * 3.76)^2 + (0.074 * 3.76)^2)} = 7.08 \mu\text{m}$
- Movement per step = $7.08 \mu\text{m} / 10 = 0.708 \mu\text{m}$

2. Centre to North:

- $\Delta X = 4664.696 - 4664.968 = -0.272$ pixels
- $\Delta Y = 3199.136 - 3200.926 = -1.79$ pixels
- Movement = $\sqrt{((-0.272 * 3.76)^2 + (-1.79 * 3.76)^2)} = 6.85 \mu\text{m}$
- Movement per step = $6.85 \mu\text{m} / 10 = 0.685 \mu\text{m}$

3. Centre to South:

- $\Delta X = 4664.742 - 4664.935 = -0.193$ pixels
- $\Delta Y = 3202.414 - 3200.614 = 1.8$ pixels
- Movement = $\sqrt{((-0.193 * 3.76)^2 + (1.8 * 3.76)^2)} = 6.77 \mu\text{m}$
- Movement per step = $6.77 \mu\text{m} / 10 = 0.677 \mu\text{m}$

4. Centre to East:

- $\Delta X = 4663.116 - 4664.924 = -1.808$ pixels
- $\Delta Y = 3200.045 - 3200.193 = -0.148$ pixels
- Movement = $\sqrt{((-1.808 * 3.76)^2 + (-0.148 * 3.76)^2)} = 6.82 \mu\text{m}$
- Movement per step = $6.82 \mu\text{m} / 10 = 0.682 \mu\text{m}$

New AO Unit:

Experiment 3: 20 Steps

1. Centre to West:

- $\Delta X = 4246.305 - 4243.739 = 2.566$ pixels
- $\Delta Y = 3127.305 - 3128.057 = -0.752$ pixels
- Movement = $\sqrt{((2.566 * 3.76)^2 + (-0.752 * 3.76)^2)} = 10.11 \mu\text{m}$
- Movement per step = $10.11 \mu\text{m} / 20 = 0.506 \mu\text{m}$

2. Centre to North:

- $\Delta X = 4243.253 - 4243.68 = -0.427$ pixels
- $\Delta Y = 3124.584 - 3127.622 = -3.038$ pixels
- Movement = $\sqrt{((-0.427 * 3.76)^2 + (-3.038 * 3.76)^2)} = 11.58 \mu\text{m}$

- Movement per step = $11.58 \mu\text{m} / 20 = 0.579 \mu\text{m}$

3. Centre to South:

- $\Delta X = 4244.157 - 4243.676 = 0.481$ pixels

- $\Delta Y = 3129.281 - 3126.815 = 2.466$ pixels

- Movement = $\sqrt{((0.481 * 3.76)^2 + (2.466 * 3.76)^2)} = 9.58 \mu\text{m}$

- Movement per step = $9.58 \mu\text{m} / 20 = 0.479 \mu\text{m}$

4. Centre to East:

- $\Delta X = 4240.781 - 4243.412 = -2.631$ pixels

- $\Delta Y = 3127.318 - 3127.37 = -0.052$ pixels

- Movement = $\sqrt{((-2.631 * 3.76)^2 + (-0.052 * 3.76)^2)} = 9.89 \mu\text{m}$

- Movement per step = $9.89 \mu\text{m} / 20 = 0.495 \mu\text{m}$

Experiment 4: 100 Steps

1. Centre to West:

- $\Delta X = 4251.762 - 4243.32 = 8.442$ pixels

- $\Delta Y = 3124.839 - 3126.364 = -1.525$ pixels

- Movement = $\sqrt{((8.442 * 3.76)^2 + (-1.525 * 3.76)^2)} = 32.80 \mu\text{m}$

- Movement per step = $32.80 \mu\text{m} / 100 = 0.328 \mu\text{m}$

2. Centre to North:

- $\Delta X = 4242.112 - 4243.51 = -1.398$ pixels

- $\Delta Y = 3116.439 - 3126.252 = -9.813$ pixels

- Movement = $\sqrt{((-1.398 * 3.76)^2 + (-9.813 * 3.76)^2)} = 38.27 \mu\text{m}$

- Movement per step = $38.27 \mu\text{m} / 100 = 0.383 \mu\text{m}$

3. Centre to South:

- $\Delta X = 4244.229 - 4242.938 = 1.291$ pixels

- $\Delta Y = 3134.987 - 3126.23 = 8.757$ pixels

- Movement = $\sqrt{((1.291 * 3.76)^2 + (8.757 * 3.76)^2)} = 33.58 \mu\text{m}$

- Movement per step = $33.58 \mu\text{m} / 100 = 0.336 \mu\text{m}$

4. Centre to East:

- $\Delta X = 4233.869 - 4243.212 = -9.343$ pixels

- $\Delta Y = 3126.851 - 3125.807 = 1.044$ pixels

- Movement = $\sqrt{((-9.343 * 3.76)^2 + (1.044 * 3.76)^2)} = 35.50 \mu\text{m}$

- Movement per step = $35.50 \mu\text{m} / 100 = 0.355 \mu\text{m}$

Experiment 5: 100 Steps

1. Centre to West:

- $\Delta X = 5746.512 - 5738.303 = 8.209$ pixels

- $\Delta Y = 3107.134 - 3107.575 = -0.441$ pixels

- Movement = $\sqrt{((8.209 * 3.76)^2 + (-0.441 * 3.76)^2)} = 31.01 \mu\text{m}$
- Movement per step = $31.01 \mu\text{m} / 100 = 0.310 \mu\text{m/step}$

2. Centre to North:

- $\Delta X = 5736.099 - 5737.808 = -1.709$ pixels
- $\Delta Y = 3098.667 - 3108.756 = -10.089$ pixels
- Movement = $\sqrt{((-1.709 * 3.76)^2 + (-10.089 * 3.76)^2)} = 38.97 \mu\text{m}$
- Movement per step = $38.97 \mu\text{m} / 100 = 0.390 \mu\text{m/step}$

3. Centre to South:

- $\Delta X = 5739.808 - 5738.341 = 1.467$ pixels
- $\Delta Y = 3116.82 - 3107.728 = 9.092$ pixels
- Movement = $\sqrt{((1.467 * 3.76)^2 + (9.092 * 3.76)^2)} = 34.76 \mu\text{m}$
- Movement per step = $34.76 \mu\text{m} / 100 = 0.348 \mu\text{m/step}$

4. Centre to East:

- $\Delta X = 5728.264 - 5737.906 = -9.642$ pixels
- $\Delta Y = 3109.682 - 3109.01 = 0.672$ pixels
- Movement = $\sqrt{((-9.642 * 3.76)^2 + (0.672 * 3.76)^2)} = 36.41 \mu\text{m}$
- Movement per step = $36.41 \mu\text{m} / 100 = 0.364 \mu\text{m/step}$

Experiment 6: 1000 Steps

1. Centre to West:

- $\Delta X = 5747.25 - 5737.757 = 9.493$ pixels
- $\Delta Y = 3108.026 - 3108.593 = -0.567$ pixels
- Movement = $\sqrt{((9.493 * 3.76)^2 + (-0.567 * 3.76)^2)} = 35.81 \mu\text{m}$
- Movement per step = $35.81 \mu\text{m} / 1000 = 0.0358 \mu\text{m/step}$

2. Centre to North:

- $\Delta X = 5736.701 - 5738.085 = -1.384$ pixels
- $\Delta Y = 3098.707 - 3109.206 = -10.499$ pixels
- Movement = $\sqrt{((-1.384 * 3.76)^2 + (-10.499 * 3.76)^2)} = 39.78 \mu\text{m}$
- Movement per step = $39.78 \mu\text{m} / 1000 = 0.0398 \mu\text{m/step}$

3. Centre to South:

- $\Delta X = 5739.036 - 5737.644 = 1.392$ pixels
- $\Delta Y = 3117.517 - 3108.561 = 8.956$ pixels
- Movement = $\sqrt{((1.392 * 3.76)^2 + (8.956 * 3.76)^2)} = 34.12 \mu\text{m}$
- Movement per step = $34.12 \mu\text{m} / 1000 = 0.0341 \mu\text{m/step}$

4. Centre to East:

- $\Delta X = 5728.616 - 5738.284 = -9.668$ pixels
- $\Delta Y = 3109.651 - 3108.778 = 0.873$ pixels
- Movement = $\sqrt{((-9.668 * 3.76)^2 + (0.873 * 3.76)^2)} = 36.60 \mu\text{m}$
- Movement per step = $36.60 \mu\text{m} / 1000 = 0.0366 \mu\text{m/step}$

Key Findings

- Movement per Step
 - For 100 steps: 0.310 - 0.390 $\mu\text{m}/\text{step}$
 - For 1000 steps: 0.0341 - 0.0398 $\mu\text{m}/\text{step}$
- Linearity: The movement per step is approximately 10 times smaller for 1000 steps compared to 100 steps, indicating good linearity in the AO unit's response.
- Directional Sensitivity: The North-South axis shows slightly larger movements compared to the East-West axis, but the difference is less pronounced than in the old AO unit.
- Precision: Demonstrates high precision, with movements as small as 0.0341 μm per step (in the 1000-step test).
- Range of Motion:
 - For 100 steps: 31.01 - 38.97 μm total movement
 - For 1000 steps: 34.12 - 39.78 μm total movement
- Stability: The center position remains relatively stable between measurements, indicating good repeatability.

These findings suggest that the new AO unit offers improved precision and consistency compared to the old unit, making it well-suited for fine adjustments in the optical system. The ability to make **sub-micron** adjustments per step in the 1000-step mode provides excellent control for precise optical alignment and correction.